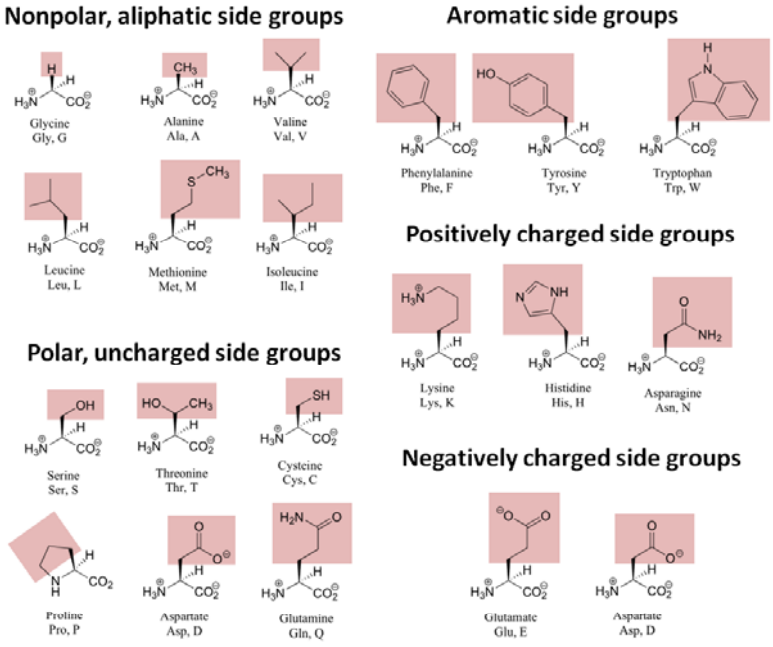
통계역학

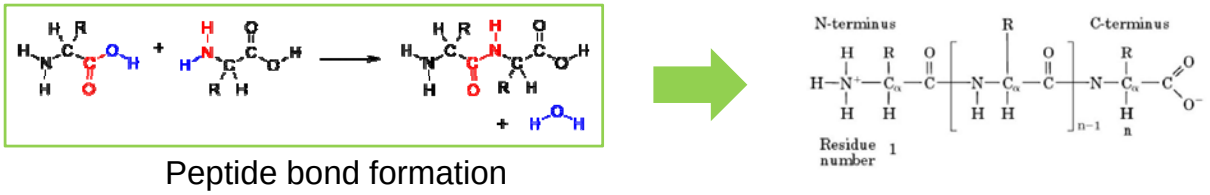
제2장 NAMD tutorial

중요 학습 내용

- **How to run NAMD**
- **Spherical boundary conditions**

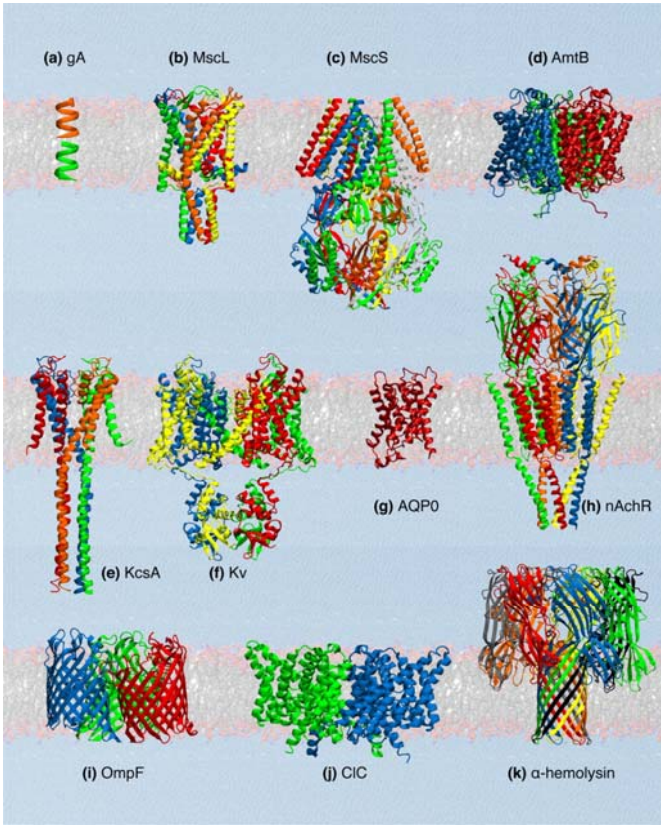
Amino acids and peptide bond formation

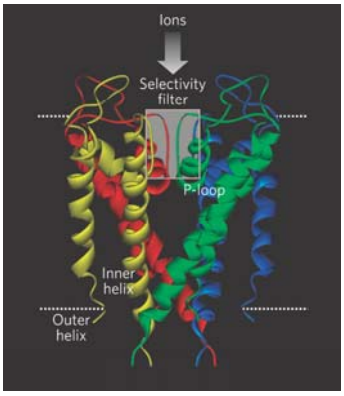


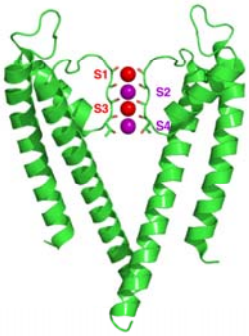


Peptide bond formation

Ion channels

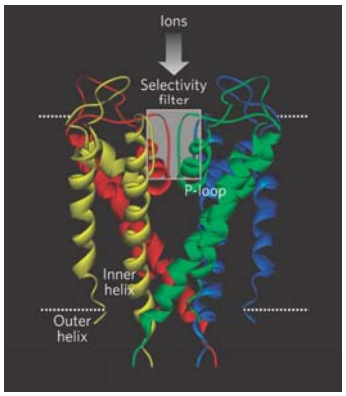






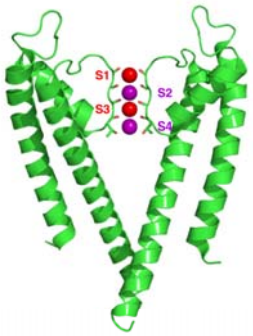
KcsA ion channel

KcsA ion channel

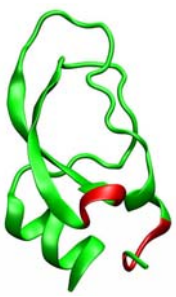




Black mamba



KcsA ion channel



Dendrotoxin

Spherical boundary conditions in NAMD

$$E_{\text{sphere}}(\mathbf{r}_i) = \begin{cases} k_{\text{sphere}}\left(|\mathbf{r}_i - \mathbf{r}_{\text{center}}| - r_{\text{sphere}}\right)^{n_{\text{sphere}}}, & \text{if } r_i > r_{\text{sphere}} \\ 0 & \text{otherwise} \end{cases}$$

$\mathbf{r}_{\text{center}}$: Center of the sphere (= sphericalBCcenter)

k_{sphere} : force constant for first potential (= sphericalBCk1)

r_{sphere} : radius for first boundary condition (= sphericalBCr1)

n_{sphere} : exponent for first boundary potential

How to make the center of mass of system the origin

In Tk Console:

```
set all [atomselect top all]
$all moveby [vecinvert [measure center $all weight mass]]
set selp [atomselect top protein]
$selp set chain A
$selp set segname UBQ
set selp [atomselect top water]
$all writepdb ubq_ws_origin.pdb
```

How to run NAMD

- Using a single core in a single node

  namd2 conf_file_name > log_file_name &

- Using multi cores in a single node (w/o CUDA)

  namd2 +p4 conf_file_name > log_file_name &

- Using multi cores in a single node with GPU (w/ CUDA)

  namd2 +idlepoll +p4 conf_file_name > log_file_name &

Root mean square deviation (RMSD)

- Definition of RMSD

1) $$RMSD_{\text{protein}}(t_j) = \sqrt{\frac{\sum_{\alpha=1}^{N_\alpha}(\mathbf{r}_\alpha(t_j) - \mathbf{r}_\alpha(0))^2}{N_\alpha}}$$

2) $$RMSD_{\text{protein}}(t_j) = \sqrt{\frac{\sum_{\alpha=1}^{N_\alpha}(\mathbf{r}_\alpha(t_j) - \langle \mathbf{r}_\alpha \rangle)^2}{N_\alpha}}$$

where

$$\langle \mathbf{r}_\alpha \rangle = \frac{1}{N_t}\sum_{j=1}^{N_t}\mathbf{r}_\alpha(t_j)$$

Maxwell-Boltzmann distributions for initial conditions

- The Maxwell-Boltzmann distribution of molecular velocities in x direction

$$f(v_x)dv_x = \left(\frac{m}{2\pi k_B T}\right)^{1/2} \exp\left[-\frac{mv_x^2}{2k_B T}\right] dv_x$$

- The Maxwell-Boltzmann distribution of molecular velocities in three dimensions

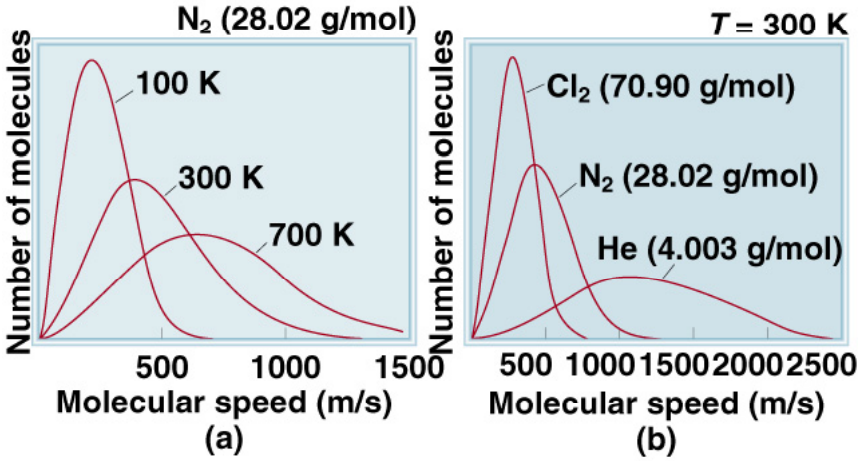
$$f(v_x) f(v_x) f(v_x) dv_x dv_y dv_z = \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right] dv_x dv_y dv_z$$

$$= \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left[-\frac{mv^2}{2k_B T}\right] dv_x dv_y dv_z$$

Maxwell distribution of speeds

- The Maxwell distribution of speeds

$$f(v)dv = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 \exp\left[-\frac{mv^2}{2k_B T}\right] dv$$



Maxwell distribution of kinetic energy

- Maxwell distribution of speeds

$$f(v)dv = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp\left[\frac{-mv^2}{2k_B T} \right] dv$$

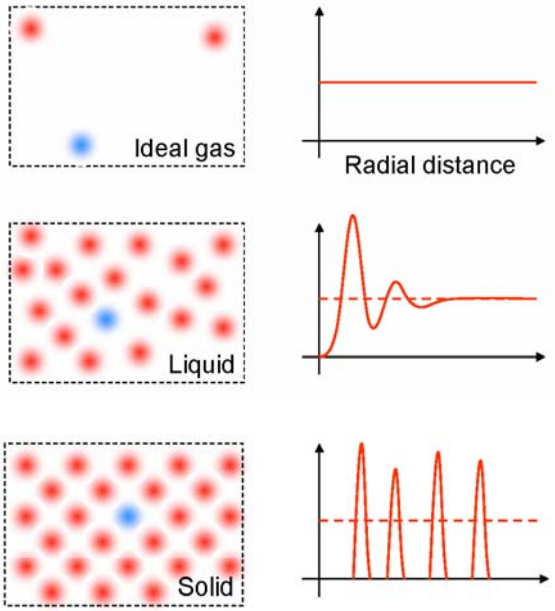
- Maxwell distribution of kinetic energy

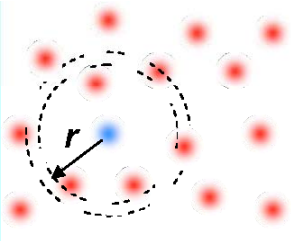
$$\epsilon_k = \frac{1}{2} m v^2 \Rightarrow dv = \left(\frac{1}{2m\epsilon_k} \right)^{1/2} d\epsilon_k$$

$$f(\epsilon_k)d\epsilon_k = \frac{2}{\sqrt{\pi}} \frac{1}{(k_B T)^{3/2}} \sqrt{\epsilon_k} \exp\left[\frac{-\epsilon_k}{k_B T} \right] d\epsilon_k$$

Radial distribution function - structured fluids

- Radial distribution function of gas, liquid, and solid





- Calculate the number of particles inside a radial segement ***(r,r+Δr)*** from one specific particle in the system.
- Average the distribution over all particles in the system.
- Divide the (average) number of particles in a segment by the segment volume.
- Finally, normalize thus obtained radial density profile with the bulk density ***N/V*** of the system.

Pressure calculation

- Virial theorem for pressure

$$P = \rho k_B T + \frac{1}{3V}\left\langle \sum_{i=1}^{N}\sum_{j>i}^{N} \mathbf{r}_{ij} \cdot \mathbf{F}_{ij} \right\rangle$$

where $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ and $\mathbf{F}_{ij}$ is the force acting on particle *i* by particle *j*.